A Deep Dive into Model Structure: Do Deep Learning Models Learn Verifiable and Transferable Hydrological Knowledge?

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Abstract

Deep learning models have demonstrated remarkable predictive skill in hydrological modeling, yet a fundamental question remains unresolved: do these models learn physically meaningful hydrological knowledge, or merely exploit statistical correlations? Addressing this question is essential for advancing scientific machine learning from prediction toward discovery. Here, we propose a circuit-based interpretability framework, Sparse Transformer-based Explainable Hydrology (STEH), that enables systematic verification of learned knowledge across model structure, behavior, and transferability. STEH introduces structured sparsification to identify compact computational circuits and integrates ablation, sensitivity, and probing analyses to evaluate whether internal representations correspond to physically meaningful hydrological processes. Across 521 catchments in the CAMELS-US dataset, we show that simplified model structures yield clearer and more interpretable decision pathways without sacrificing predictive performance. Importantly, we demonstrate that deep learning models encode verifiable hydrological signals, including antecedent storage effects and precipitation-runoff relationships, which remain consistent under controlled perturbations.Furthermore, we find that the extracted structural knowledge is transferable across model classes, improving the performance of a Random Forest benchmark, suggesting that learned representations capture model-independent hydrological information. Our results reveal that deep learning models can internalize physically meaningful and transferable knowledge, bridging the gap between predictive accuracy and scientific interpretability. This study establishes a verification-oriented paradigm for interpretable hydrological modeling and provides a pathway toward using artificial intelligence as a tool for scientific discovery in Earth system sciences.

Keywords: Model interpretability, Explainable artificial intelligence, Structural sparsity, Hydrological modeling, Streamflow prediction, Transformer

1 Introduction

Streamflow forecasting plays a central role in water resources management, flood mitigation, and basin regulation and control[1]. In recent years, deep learning techniques have achieved rapid progress in hydrological prediction[2], providing powerful predictive capabilities for streamflow forecasting[3, 4]. However, a critical question remains unresolved: do these models learn physically meaningful feature representations, or do they rely solely on complex statistical correlations to make predictions[5]? This distinction is of decisive importance for hydrological research. If models merely achieve statistical fitting, their generalization ability will be confined to the distribution of historical data, limiting their capacity to advance the understanding of the water cycle or support discoveries in Earth science. Conversely, if models are capable of encoding verifiable hydrological knowledge, deep learning can evolve from a mere fitting tool into a driver of scientific discovery[6], enabling the identification of previously unresolved mechanisms in complex basins.

To address the "black-box" nature of deep learning hydrological models, constraint-based approaches represented by Physics-Informed Neural Networks (PINNs)[7] attempt to embed hydrological governing equations into the loss function[8], thereby enforcing physical consistency in model outputs[9]. Although this result-oriented integration of physical knowledge improves the plausibility of predictions to some extent, it does not fundamentally resolve the core issue of model interpretability[10]. Even when model outputs conform to physical laws, the internal mechanisms by which input variables are utilized and features evolve remain hidden within a complex parameter space[11]. This implies that PINNs primarily mimic physical laws rather than actively discover hydrological processes[12], and thus fail to reveal what hydrological mechanisms the model has truly learned. Consequently, their potential as tools for deep decision support and scientific knowledge discovery remains limited.

Another mainstream line of research investigates model interpretability through post-hoc analysis techniques, among which Local Interpretable Model-agnostic Explanations (LIME)[13] and SHapley Additive exPlanations (SHAP)[14] are the most widely used. These methods quantify the contributions of key hydrological variables—such as precipitation and evapotranspiration—to prediction outcomes based on cooperative game theory or local surrogate models[15], thereby breaking the "black-box" barrier in a formal sense. However, such explanations are essentially post-hoc in nature. Their results heavily depend on the statistical properties of the data distribution rather than the model's intrinsic computational logic. More critically, they treat hydrological variables as isolated features, neglecting the prevalent collinearity among variables in natural systems. This mathematical decomposition, detached from the model's internal operations, often leads to unverifiable explanations and spurious correlations, making them unreliable for inferring hydrological scientific laws or understanding streamflow generation mechanisms in high-risk basins.

Overall, both PINNs and post-hoc interpretability techniques have made significant progress in improving the interpretability of streamflow forecasting models, offering feasible pathways to mitigate the "black-box" issue. Nevertheless, a fundamental scientific gap persists in the application of deep learning to hydrology: are the feature relationships extracted by models merely incidental statistical artifacts, or do they reflect robust hydrological physical laws? This gap directly hinders the paradigm shift of deep learning from a predictive tool to a scientific discovery instrument. Specifically, bridging this gap requires overcoming three interrelated systemic limitations in current interpretability research:

First, interpretability results lack objective verifiability[16], casting doubt on the validity of scientific discoveries derived from them. Unlike streamflow prediction results, which can be directly evaluated against observed ground truth using quantitative metrics, feature attribution-based interpretations cannot be directly validated and are often justified indirectly through high predictive accuracy[17]. However, even highly accurate predictions may arise from complex but spurious statistical correlations[18], rather than physically meaningful hydrological knowledge. Without independent and quantitative validation of attribution results, it is difficult to determine whether models have truly learned physically meaningful relationships[19]. This severely limits the credibility of interpretability in high-stakes decision-making and impedes its role in scientific discovery.

Second, the lack of systematic evaluation of model behavior leads to a disconnect between interpretability results and actual hydrological physical logic. Current interpretability analyses often remain at the level of static feature importance rankings[20, 21], neglecting the need to assess behavioral consistency under dynamic hydrological conditions[22]. A scientifically meaningful hydrological model should respond to key drivers such as precipitation and evapotranspiration in accordance with fundamental hydrological principles[23]. For instance, a significant increase in precipitation should trigger a consistent rise in streamflow within the model[24]. Without such perturbation experiments or counterfactual validation[25, 26], interpretability results cannot be effectively linked to real hydrological behavior[27], limiting their value for scientific understanding.

Finally, the lack of transferability validation across model architectures hinders the extraction of universal hydrological laws. The essence of scientific discovery lies in identifying generalizable principles that remain consistent across different observational tools or modeling approaches[28]. However, existing interpretability studies are often constrained by the parameterization of specific model architectures and rarely examine whether the identified hydrological relationships hold across different model families. If the extracted representations cannot be reproduced across models and scenarios, they are more likely to reflect overfitting to specific data distributions rather than true hydrological laws[29], thereby limiting their capacity to support genuine scientific discovery.

Notably, in other scientific domains, systematic interpretability research paradigms have gradually matured, with standardized validation frameworks significantly enhancing the credibility of model explanations [30]. However, such efforts remain relatively limited in hydrological modeling, and a unified and mature interpretability validation framework has yet to be established. The absence of such standardized validation paradigms has become the final critical barrier preventing deep learning-based hydrological models from evolving from predictive tools into engines of scientific discovery.

This study aims to transcend the descriptive limitations of traditional interpretability analysis by constructing a full-chain interpretability validation framework to systematically investigate the internal information propagation mechanisms of deep learning-based hydrological models. Rather than merely focusing on *what the model predicts*, this work seeks to explore, at a fundamental level, *how information flows within the model*, and *how such internal knowledge supports model generalization under complex hydrological conditions*. To this end, the study focuses on the following four core scientific questions:

1. **Structure-performance mapping mechanism:** What intrinsic relationships exist between model structure and predictive performance?

2. **Existence and distribution of physical knowledge:** Do deep learning-based hydrological models encode physical knowledge, and if so, what are its spatial and feature-level distribution patterns?

3. **Response consistency across hydrological scenarios:** Can the model generate physically consistent feature representations under different hydrological conditions?

4. **Generality and transferability of scientific knowledge:** Is the physical knowledge learned by the model transferable across different scenarios and model architectures?

To address these scientific questions, this paper proposes an interpretable hydrological model (STEH), aiming to achieve full-chain interpretability from input features, internal model structures, to output results. The main contribution of this study lies not only in the development of an interpretable deep learning framework, but also the establishment of a reference path for hydrological model interpretability verification. Framed as ***A Deep Dive into Model Structure***, this work integrates structural analysis, behavioral verification, and cross-model evaluation to move interpretability beyond descriptive attribution toward rigorous assessment of whether model knowledge is ***verifiable***, physically meaningful, and ***transferable***. In doing so, it provides preliminary insights into a fundamental question: ***what kind of hydrological knowledge is learned, and is it verifiable and transferable?***

2 Results and discussion

Next, we conduct an interpretability study of deep learning-based streamflow prediction models using the CAMELS-US dataset. After screening, 521 study sites are selected (details are provided in the Section 4.3), and the model performs well in the majority of regions (Supplementary Information 1). By integrating the model circuitry with various interpretability methods and perspectives, we provide an in-depth explanation of how the proposed interpretable deep learning framework (STEH) better captures the internal knowledge representation and physical consistency issues that existing models struggle to resolve.

2.1 Interpretable Circuit Structures Across Hydrological Stations

Given the large number of stations (521), it is impractical to visualize the learned circuits for all stations individually. Therefore, a representative sampling strategy is adopted to systematically examine circuit structures under different predictive conditions.

Specifically, stations are categorized into three performance groups[31] based on the NSE. One representative station is selected from each category, as summarized in Table 1.

Table 1 Representative stations across NSE performance categories

Performance Category	NSE Range	Station ID	NSE
Very Good	NSE $>$ 0.75	09066300	0.9905
Good	0.65 $<$ NSE $\leq$ 0.75	01439500	0.7366
Satisfactory	0.50 $<$ NSE $\leq$ 0.65	03504000	0.6217

We first consider a representative inland station with very good performance (Station ID: 09066300, *Middle Creek near Minturn, CO.*, 39.65°N, 106.38°W, NSE = 0.9905) to illustrate the extracted core circuit (Fig. 1a). The station is located in a high-altitude inland basin in Colorado, USA (approximately 3251 m), where hydrological processes are strongly influenced by temperature-driven dynamics and catchment routing[32].

The circuit structure (Fig. 1a) shows that the model retains a clear and stable information flow pathway after strong sparsification. Among the inputs, streamflow (Q) and maximum temperature (Tmax) dominate, with substantially stronger connections than DayL. Beyond variable importance, this structure suggests a process-level pattern in which antecedent catchment state (with Q acting as a storage proxy) and temperature jointly regulate runoff release. The recovered pathway is consistent with plausible high-altitude controls such as snowmelt timing and evapotranspiration sensitivity[33]. Within the hidden layers, information propagates through a small number of key nodes and converges to the output, forming a compact, low-redundancy computational pathway.

Training dynamics (Fig. 1d, lower right) provide complementary evidence: as the number of retained nodes (K) increases, NSE rapidly improves to near 1 and stabilizes when K remains relatively small (K ¡ 100), while MSE decreases sharply and stays low. Notably, performance comparable to the full model is achieved with only a small subset of nodes. This indicates that the extracted sparse circuit preserves essential predictive capability, supporting its functional fidelity[34]. More broadly, the sparsity-inducing training appears to retain critical computational units while removing redundant structure[35].

Furthermore, the interaction analysis (Fig. 1e) reveals strong dependencies among input variables. In particular, Tmax exhibits the strongest interaction with Q, followed by the interaction between Q and DayL, while DayL shows relatively weak interactions overall. This supports a process-level interpretation: thermal forcing does not act independently, but modulates discharge response through antecedent-state conditioning, consistent with energy-state coupling in mountain basins[36].

Finally, the contribution analysis (Fig. 1f) shows a clear imbalance among input variables. Streamflow (Q) contributes the most, followed by Tmax, while DayL has the smallest contribution. Combined with the circuit and interaction evidence, this indicates that the dominant hydrological pattern at this site is a state-dependent runoff pathway with secondary temperature regulation, rather than a uniformly distributed multi-driver response.

Overall, for this very good inland station, the extracted core circuit is highly sparse while maintaining strong predictive capability[37]. More importantly, STEH recovers a coherent hydrological mode in which antecedent state and thermal forcing jointly govern streamflow response, suggesting that the learned sparse circuit reflects hydrologically meaningful structure rather than only statistical attribution.

To test whether this pattern persists beyond the best case, we next examine a representative station with good performance (Station ID: 01439500, *Bush Kill at Shoemakers, PA*, 41.09°N, 75.04°W, NSE = 0.7366). The extracted circuit exhibits a moderately sparse structure (Fig. 1c). Compared with the very good inland station, a broader set of inputs contribute to prediction, including precipitation (P), streamflow (Q), and radiation-related variables. The resulting information flow is therefore more distributed[35], consistent with multiple sub-processes acting simultaneously. Interaction analysis still shows a dominant role for Q, but with stronger coupling to precipitation, which aligns with an event-response pathway where rainfall signals are filtered by antecedent wetness before being transmitted to discharge[38].

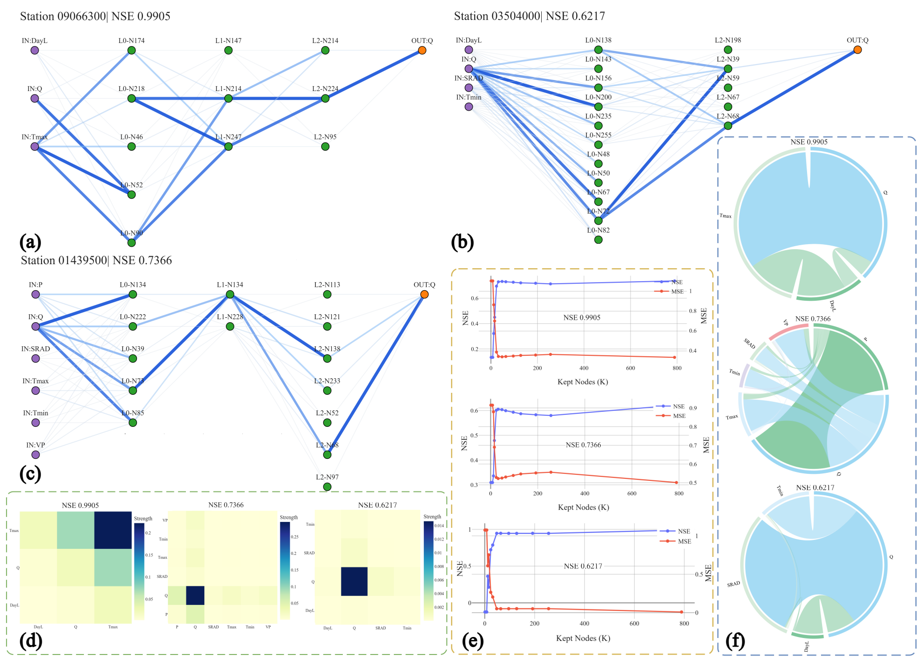

Fig. 1 **Sparse Circuit Extraction and Interpretability Analysis of Hydrological Forecasting Models**(a), (b), (c) Core Circuit Structures: Deep learning model circuits for high-performance (NSE = 0.9905), medium-performance (NSE = 0.7366), and low-performance (NSE = 0.6217) sites, showing sparse connectivity from input to output layers.(d) Input Variable Interaction Heatmap: Pairwise dependencies between input variables, with darker colors indicating stronger interactions.(e) Node Pruning Training Curves: Performance and error evolution as node count changes, revealing model redundancy and extraction effectiveness.(f) Input Variable Contribution Diagram: Visualization of input variable contributions to flow prediction, with chord width indicating influence magnitude.*In panel (d), performance decreases from left to right; in panels (e) and (f), top to bottom represents performance from optimal to worst.*

When performance decreases further, the learned structure becomes visibly less coherent. For a representative station with satisfactory performance (Station ID: 03504000, *Nantahala River near Rainbow Springs, NC*, 35.13°N, 83.62°W, NSE = 0.6217), the extracted circuit is less structured and more diffuse (Fig. 1b). Although streamflow (Q) still dominates prediction, other variables such as radiation (SRAD) and DayL show weaker and less consistent contributions. Interaction patterns are also less pronounced, suggesting that no single robust hydrological pathway is cleanly separated from competing signals. The circuit retains more redundant connections, and sparsity-inducing training appears less effective in isolating a compact predictive core[33].

In our comparative analysis, we observed an interesting phenomenon: as prediction performance decreases, the sparsity of the extracted circuits also decreases, and the structure tends to become more dispersed. Specifically, high-performance sites exhibit compact and orderly pathways, which correspond to clear hydrological patterns, while low-performance sites show increased redundancy, weaker interaction consistency, and greater instability in the interpretability of the knowledge layer[39]. This trend has attracted our attention. As a result, we conducted a more systematic analysis to verify *whether there is a significant and stable relationship between the degree of model structure simplification and prediction performance*.

2.2 Parsimony and Physical Interpretability Are Prevalent

Interestingly, our findings reveal a trend that contrasts with observations reported in some prior studies[40, 41], where increased model complexity has been associated with improved predictive performance. As shown in Table 2, models with higher predictive skill (NSE $\geq$ 0.5) exhibit significantly lower structural complexity compared to lower-performing counterparts, as indicated by a reduced number of nodes (45.77 vs. 56.35, $p = 5.00 \times 10^{-5}$). This suggests that increased model complexity does not necessarily translate into improved performance in this context; instead, more parsimonious architectures may facilitate better generalization[39]. One possible explanation is that overly complex models introduce redundant representations or noise, which can degrade predictive robustness in hydrological applications[42].

Table 2 Structural complexity across NSE-based performance groups

Group	Stations	Mean Nodes	Median Nodes
NSE $<$ 0.5	194	56.35	56.50
NSE $\geq$ 0.5	327	45.77	42.00
All	521	49.71	49.00

Note: A statistically significant difference is observed between groups ($p = 5.00 \times 10^{-5}$).

From the perspective of physical interpretability, the probe analysis (Table 3) indicates that key hydrological variables—including CUM, API, and CHG—are consistently encoded in the learned representations across models[43]. Specifically, no statistically significant differences are observed in probe performance between NSE-based groups (all $p > 0.05$). This implies that the ability to capture physically meaningful signals is broadly preserved across model groups[18], rather than being exclusive to high-performing models.

Table 3 Probe performance under different NSE conditions

Probe	Group	Mean	Median	p-value
R^2_{cum}	NSE < 0.5	0.8746	0.9639	0.4341
	NSE ≥ 0.5	0.8915	0.9617	
	All	0.8851	0.9621	
R^2_{api}	NSE < 0.5	0.8795	0.9682	0.5676
	NSE ≥ 0.5	0.8917	0.9651	
	All	0.8871	0.9663	
R^2_{chg}	NSE < 0.5	0.3791	0.4267	0.1563
	NSE ≥ 0.5	0.4054	0.4406	
	All	0.3955	0.4342	

Outliers were excluded using the 1.5 IQR criterion within each group. Results remain consistent without filtering.

Taken together, these results suggest that parsimony and physical interpretability are not competing objectives. Models with simpler structures can still achieve strong predictive performance while encoding essential hydrological knowledge related to streamflow generation. This naturally motivates the next question: *if such knowledge is broadly present across models, what is its concrete content and spatial organization across basins?*

2.3 Core Circuits Reflect Hydrological Knowledge Across Basins

To answer that question, we examine the spatial organization of the learned interaction structures within core circuits. The results indicate that the model captures structured and regime-dependent hydrological knowledge across diverse basins[44]. As shown in Fig. 2(a), the dominant interaction in most basins is between precipitation and discharge (P–Q). This pattern reflects the fundamental role of precipitation as the primary driver of streamflow generation[45]. The detailed procedure for how we performed the classification can be found in the Supplementary Information 3.1.

In contrast, a subset of basins is characterized by strong interactions between lagged discharge and target discharge (Q–$\mathrm{Q}_{\mathrm{target}}$), suggesting pronounced memory effects[3]. In these basins, streamflow dynamics are more strongly influenced by antecedent conditions than by immediate precipitation inputs, which is typically associated with limited precipitation or strong storage capacity. This pattern can be interpreted as a storage-controlled response mode, where antecedent wetness and catchment retention regulate subsequent flow release.

Temperature-related interactions ($\mathrm{T}_{\mathrm{max}}$–Q and $\mathrm{T}_{\mathrm{min}}$–Q) are primarily observed in inland and high-elevation regions. These patterns indicate that temperature plays a

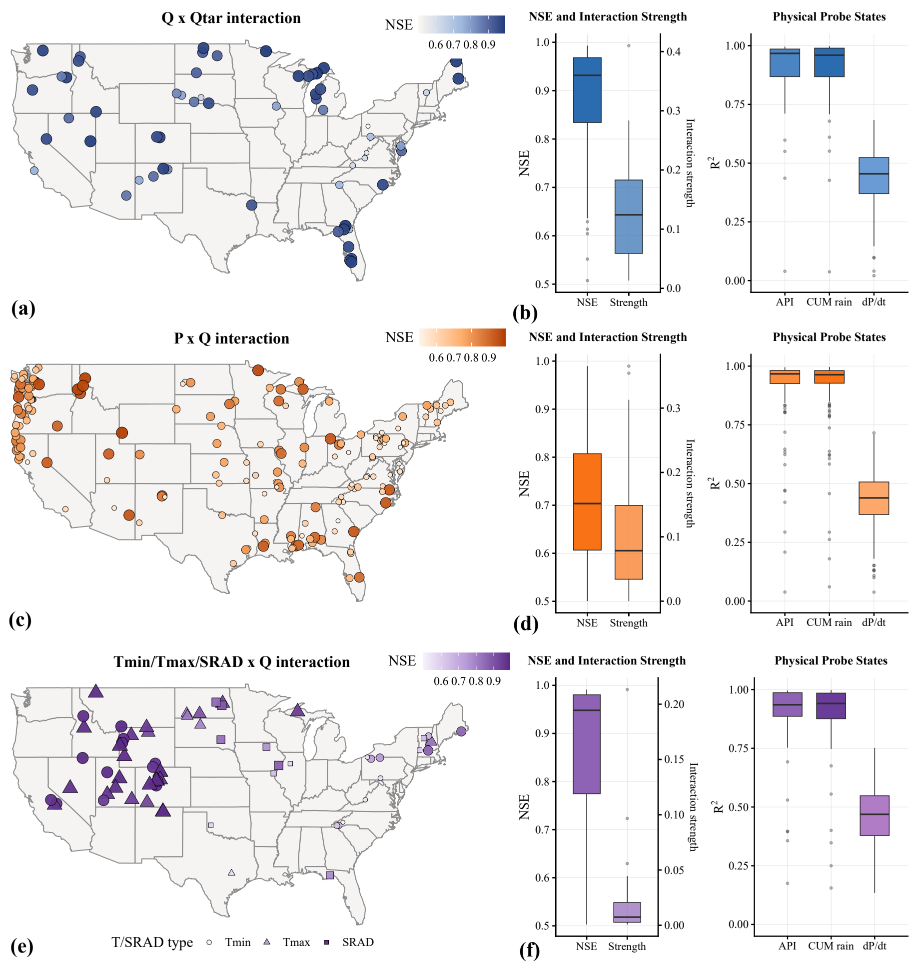

Fig. 2 Spatial patterns of hydrological interaction regimes and their performance (a,c,e) Spatial distribution of dominant interaction modes (e.g., P–Q, Q–$Q_{\mathtt{target}}$, and energy-related). (b,d,f) NSE, interaction strength, and physical probe states for the corresponding interaction types.

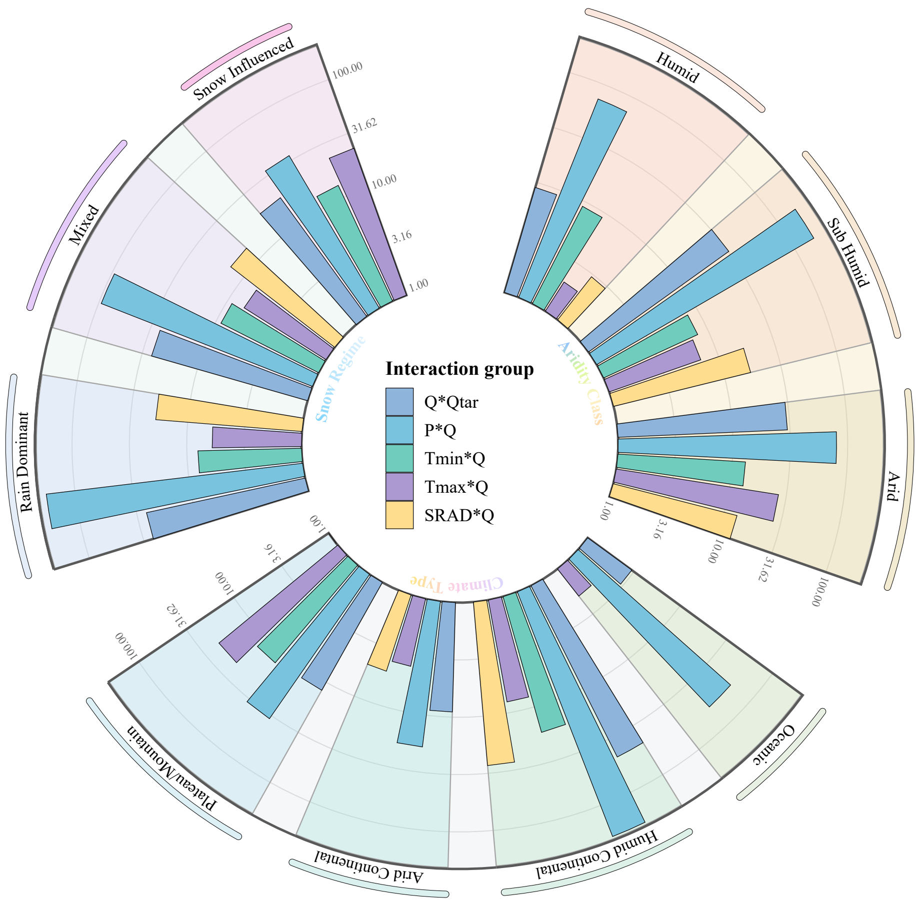

Fig. 3 Distribution of dominant interaction groups across different hydro-climatic regimes, including aridity levels, snow influence, and geo-climatic types.

critical role in regulating streamflow generation in such environments, not only through snowmelt processes but also via evaporation and other energy-limited controls[2]. In addition, interactions involving shortwave radiation (SRD) emerge in some basins, further highlighting the importance of energy inputs[46] in modulating hydrological responses. Together, these signals correspond to an energy-constrained runoff mode, distinct from precipitation-dominated systems.

These patterns remain consistent across different hydro-climatic conditions, as illustrated in Fig. 2(b–e). Precipitation-driven interactions dominate across most basins, whereas temperature-related and lagged discharge interactions become more prominent under specific environmental settings. In arid and semi-humid regions, temperature and memory-related interactions are more frequently observed[2], indicating stronger energy constraints and persistence effects. In contrast, humid basins are largely governed by precipitation-driven dynamics[45].

Similarly, temperature-related interactions[46] are concentrated in snow-influenced basins, consistent with snowmelt-controlled streamflow processes, while precipitation-driven interactions dominate in rain-fed systems[47]. Across different geo-climatic types, distinct interaction structures exhibit clear regional preferences, suggesting that the learned circuit structures adapt to local environmental conditions.

Overall, these findings indicate that the learned core circuits are not arbitrary but systematically reflect regime-dependent hydrological patterns[48]. Thus, STEH moves the interpretation target from "which variable is important" to "which process pathway is operating," providing structural evidence that the model internalizes physically meaningful knowledge beyond simple variable associations. However, structural coherence alone is insufficient: the extracted knowledge should also produce physically consistent behavior under controlled perturbations.

2.4 Accurate Volume Response but Limited Sensitivity to Temporal and State Perturbations

Following the structural analysis above, we next test whether the extracted hydrological knowledge corresponds to realistic model behavior. As shown in Fig. 4, the counterfactual analysis indicates that the model exhibits physically meaningful responses under different types of perturbations[27]. For magnitude-based interventions, particularly rainfall scaling (Fig. 4b), the model produces directionally consistent and approximately linear changes in streamflow volume across stations. As precipitation increases (or decreases), the predicted streamflow volume correspondingly increases (or decreases), while prediction error remains largely unchanged. This indicates that the model has learned a stable and physically consistent mapping between precipitation and streamflow generation, and can respond to changes in input water magnitude without substantial loss of predictive accuracy[24].

Fig. 4 Counterfactual sensitivity analysis across intervention types. (a) Station-level expected behavior statistics. (b) Rainfall scaling. (c) Rainfall pattern redistribution. (d) streamflow-state scaling. (e) Temperature perturbation. Blue lines denote relative changes in predicted streamflow volume, and orange lines denote relative changes in prediction error. Detailed experimental design and evaluation metrics are provided in Supplementary Information 3.2.

Under rainfall pattern redistribution (Fig. 4c), the total precipitation is conserved, and the predicted streamflow volume remains broadly stable, consistent with mass conservation expectations[49]. Structural differences nevertheless emerge across redistribution modes. In particular, the back-loaded configuration shows a clear positive deviation in predicted streamflow volume relative to other scenarios. This indicates that the model is sensitive to the temporal concentration of rainfall and produces stronger streamflow responses[44] when precipitation is concentrated toward later periods, reflecting an initial capability to capture rainfall-streamflow lag relationships. Meanwhile, the variation in prediction error across scenarios indicates that the model's utilization of temporal structure remains somewhat scenario-dependent.

For streamflow-state perturbations (Fig. 4d), moderate scaling of recent streamflow history leads to corresponding adjustments in predicted volume, while prediction error remains stable. This indicates that the model can utilize state information while maintaining consistent predictive performance. As the scaling magnitude increases, however, the model response gradually saturates, with volume changes no longer increasing proportionally to the input. This attenuation pattern suggests that the model does not simply amplify streamflow magnitude linearly, but may impose constraints on non-structured or inconsistent input variations, thereby avoiding unrealistic over-amplification. Such behavior is consistent with a model that captures streamflow dynamics through interactions among multiple hydrological factors[50] rather than relying solely on absolute magnitude.

For temperature perturbations (Fig. 4e), the predicted streamflow volume shows a slight decreasing trend, with consistent direction across snow-dominated and rainfall-dominated catchments and no significant difference between groups. This aligns with hydrological expectations that warming generally reduces streamflow through enhanced evapotranspiration or phase-related processes. In contrast, prediction error exhibits different sensitivities between the two regimes: snow-dominated catchments show substantially stronger responses to temperature changes than rainfall-dominated ones[2], and this difference is consistent across warming magnitudes ($p < 0.001$). This indicates that the model not only captures the overall volumetric effects of temperature, but also differentiates between distinct hydrological knowledge patterns, reflecting sensitivity to process heterogeneity.

Overall, the model demonstrates robust responses to precipitation-driven volume changes and selective, generally limited sensitivity to temporal structure and catchment state variations[51], indicating effective learning of key hydrological knowledge with remaining gaps in temporal-state representation. At the same time, the differing response patterns across perturbation types suggest that the representation of more complex processes, such as temporal organization and state evolution, remains an area for further improvement. This behavioral consistency supports the validity of the extracted knowledge, and naturally leads to the final question: *can this knowledge be transferred to improve other model classes?*

2.5 Cross-Model Transfer Validates STEH-Derived Knowledge

To answer this final question, we evaluate transferability using an external model family. Specifically, we designed a comparative experiment in which STEH-derived structural information is incorporated into a conventional Random Forest model[52] and compared against a baseline model without such information. The key purpose is not only to improve another model, but to test whether the extracted structures reflect model-independent hydrological knowledge that remains effective outside the Transformer architecture[53].

Furthermore, station-wise paired tests (paired t-test and Wilcoxon signed-rank test) reported in Supplementary Information 3.3 indicate that the modified model outperforms the baseline with statistical significance, supporting the robustness of the observed improvements.

From an overall perspective, the modified model exhibits consistent performance improvements across stations[54]. Specifically, NSE increases from 0.760 to 0.767, accompanied by reductions in both MSE and MAE, indicating improved predictive accuracy and error control[55]. The station-wise comparison (Fig. 5(d)) shows that most points lie above the diagonal line, and the gains persist across a wide range of baseline performance levels (Fig. 5(e)). These observations indicate that the transferred structures retain predictive value in an independent model class, consistent with the claim that STEH extracts hydrological knowledge beyond architecture-specific heuristics[56], while the magnitude of improvement remains modest at the aggregate level.

Table 4 Performance comparison between baseline and STEH-informed modified Random Forest models across hydro-climatic groups

Model	Group	MSE	MAE	NSE
Baseline	Arid	0.140054	0.260355	0.841540
	Humid	0.324556	0.435797	0.700072
	Sub-humid	0.270571	0.385097	0.724154
	All Stations	0.237974	0.353715	0.759683
Modified	Arid	0.133525	0.248404	0.846403
	Humid	0.313798	0.425727	0.709464
	Sub-humid	0.259417	0.372654	0.733143
	All Stations	0.228561	0.342049	0.767310

All metrics are averaged across stations within each hydro-climatic group. Only stations with STEH model performance (NSE > 0.5) are included, as the extracted hydrological structures are derived from the STEH model and may become unreliable when its predictive performance is low.

Across different hydro-climatic regimes, the most pronounced improvements are observed in arid regions. As shown in Fig. 5(a) and Table 4, inland arid and semi-arid areas exhibit predominantly positive performance gains. In the Arid group, the modified model consistently outperforms the baseline in terms of MSE, MAE, and NSE, suggesting that incorporating structural information can enhance predictive capability under such conditions. This is consistent with the intuition that in environments where precipitation signals are weak and hydrological responses are governed by multiple interacting factors[57], additional structural features can help the model better capture streamflow dynamics.

In humid regions, the modified model also achieves consistent improvements (NSE:

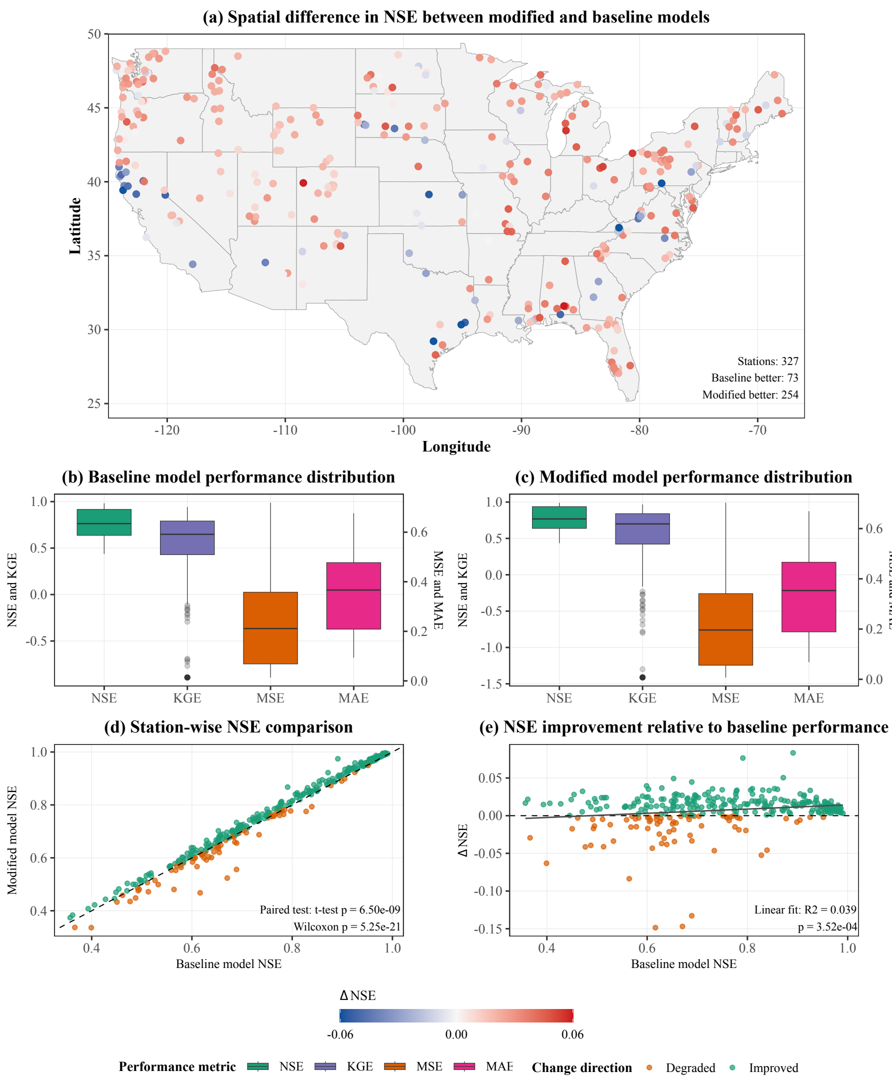

Fig. 5 **Performance comparison between baseline and STEH-informed Random Forest models.** (a) Spatial distribution of NSE differences (red: improved; blue: degraded). (b,c) Performance distributions of the baseline and modified models. (d) Station-wise NSE comparison (points above the diagonal indicate improvement). (e) NSE improvement relative to baseline performance. All results are based on stations with STEH model performance (NSE > 0.5).

0.700 to 0.709), although the magnitude of improvement is relatively smaller. This can be attributed to the comparatively simpler hydrological processes in such regions[58], where streamflow generation is primarily controlled by a dominant driver. Under these conditions, the baseline model is already capable of capturing the main dynamics, leaving limited room for further improvement. Nevertheless, the continued reduction in error metrics indicates that structural information still contributes to improved fine-scale representation[59].

Overall, these findings demonstrate that the structural information extracted by STEH is not only interpretable but also scientifically verifiable through cross-model transfer. In particular, in regions characterized by complex or multi-factor-controlled processes, incorporating such structural knowledge can substantially improve model performance. Successful transfer to Random Forest provides an external validation step: the extracted knowledge is not tied to a single network design, but reflects portable hydrological regularities. In this section we successfully illustrate several major questions: hydrological knowledge is broadly encoded across models, can be structurally identified, can be behaviorally validated for physical consistency, and can be externally verified through cross-model transfer.

2.6 Limitations

Nevertheless, despite the progress made in uncovering knowledge representations in deep learning–based hydrological models, several limitations should be acknowledged.

First, although STEH improves structural interpretability, its results still cannot be interpreted as strict causal relationships [60]. This limitation arises from the intrinsic nature of deep learning models as statistical learning systems [61], where extracted patterns primarily reflect statistical associations rather than true physical causality. Therefore, the findings should be interpreted with caution when relating them to real-world physical processes.

Second, although this study reveals interaction structures among variables, their theoretical linkage to explicit physical process equations remains to be further explored. For example, the specific coupling relationships between variables, the pathways through which interactions operate, and the development of more controllable and interpretable circuit structures are not yet fully characterized [62]. Moreover, although the current framework enforces sparsity to simplify internal structures, the analysis is still largely limited to the relationship between input variables and the final hidden representations, leaving the information evolution within intermediate high-dimensional feature spaces insufficiently explored.

In addition, the validation of the extracted "physical knowledge" in this study is still based on machine learning models rather than explicit process-based hydrological models [63]. This may limit the physical rigor of the conclusions, and future work should incorporate process-driven models to further verify the findings.

3 Summary

Deep learning has advanced hydrological prediction, but model interpretability remains challenged by low verifiability, neglected variable dependencies[18], and unclear internal structures[64], leaving uncertain whether high-performance models learn physically meaningful hydrological processes[65]. This study proposes the Sparse Transformer-based Explainable Hydrology (STEH) framework. It establishes an end-to-end interpretability pipeline using structured sparsification, circuit-level ablation, variable interaction analysis[35], and representation probing to systematically reveal the internal decision-making mechanisms of deep learning models in hydrology.

The main findings are as follows.First, STEH provides a circuit-level view to explicitly identify decision pathways and information flow, offering a structurally grounded and verifiable interpretability paradigm.Second, simpler models form more compact internal circuits, while over-complex models introduce redundant structures; model parsimony improves clarity while preserving physical meaning.Third, physically meaningful representations are learned across models regardless of performance, indicating physical knowledge is inherent in data-driven models.Fourth, STEH reliably characterizes stable and hydrologically consistent variable interactions across diverse hydroclimatic regimes.Fifth, sensitivity analysis shows the model effectively captures dominant streamflow-generation knowledge but incompletely represents temporal organization and state evolution.Sixth, STEH-derived knowledge is transferable and practically useful, as validated by performance improvements in a Random Forest model.Seventh, this study provides a replicable pathway for interpretability and physical validation of deep learning models in hydrological research.

In summary, this study demonstrates that hydrological deep learning models encode physically meaningful information that can be systematically uncovered through structural analysis. The STEH framework bridges black-box prediction and hydrological interpretation, supports knowledge discovery, and offers a generalizable route for integrating data-driven and physics-based hydrological modeling.

4 Methods and data

4.1 Framework Overview

The overall workflow of the proposed Sparse Transformer-based Explainable Hydrology (STEH) framework is illustrated in Fig. 6. The framework provides an end-to-end interpretability pipeline for deep learning-based streamflow prediction, enabling systematic analysis from hydrological input variables, through internal circuit structures, to prediction outputs.

The STEH framework consists of three main components: (1) sparse circuit discovery, (2) circuit verification and refinement, and (3) physical interpretation. Together, these components aim to reveal how deep learning models process hydrological information and whether the learned representations correspond to meaningful hydrological processes.Terminology follows the Appendix 1 terminology table (Table 5).

First, sparse circuit discovery is performed by introducing learnable gating mechanisms into the Transformer architecture. Gates are applied to input variables, attention heads, and feed-forward neurons, allowing the model to automatically identify a compact set of critical computational units during training. Through sparsity-inducing optimization, redundant structures are gradually suppressed, resulting in a sparse computational circuit responsible for streamflow prediction.

Second, circuit verification and refinement are conducted through systematic ablation experiments. Individual computational units—including input nodes, attention heads, and neurons—are selectively removed to measure their structural contribution to prediction performance. Units with negligible influence are discarded, yielding a compact core predictive circuit that preserves the essential functional structure of the model.

Third, the extracted core circuit is further analyzed to understand internal information flow and structural dependencies. Interaction analysis is used to identify cooperative or redundant relationships between model components. In addition, linear probing techniques are applied to hidden representations to evaluate whether internal states encode physically meaningful hydrological variables.

Finally, the interpretability results are validated from a hydrological perspective. Based on the structural knowledge extracted from the neural network, modifications are introduced to a Random Forest benchmark model to examine whether the discovered hydrological knowledge can improve data-driven hydrological simulations.

In what follows, we first introduce the sparse Transformer architecture and the three-stage sparsification strategy, and then describe the interpretability analyses and evaluation metrics.

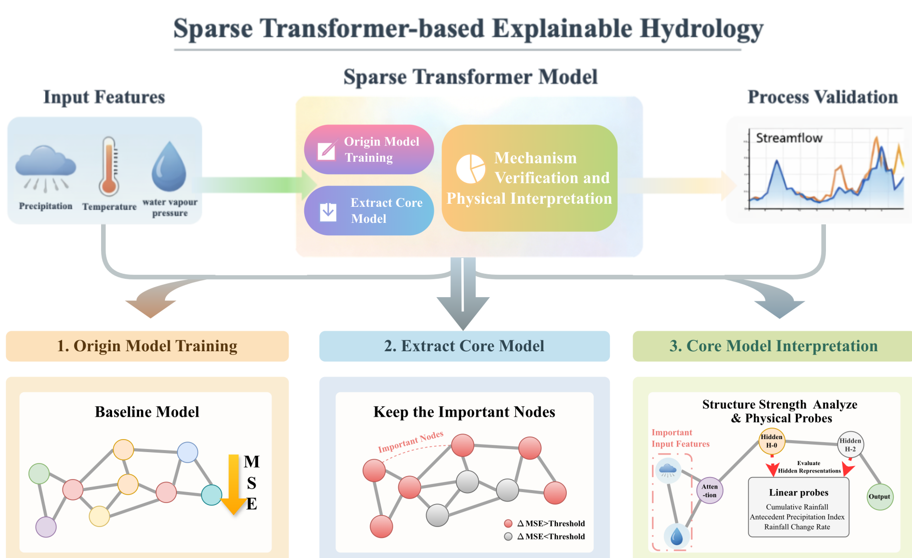

Fig. 6 Overall architecture of the proposed STEH framework

4.2 Sparse Transformer

The Transformer architecture has recently been introduced into hydrological time-series modeling because its self-attention mechanism can effectively capture long-range temporal dependencies in hydro-meteorological sequences[66]. In streamflow prediction, antecedent rainfall, temperature, and soil moisture conditions jointly influence discharge responses across multiple time lags, making long-term dependency modeling particularly important[54]. The Transformer assigns different weights to each input feature through the self-attention mechanism, automatically learning the relationships between different time steps[67].

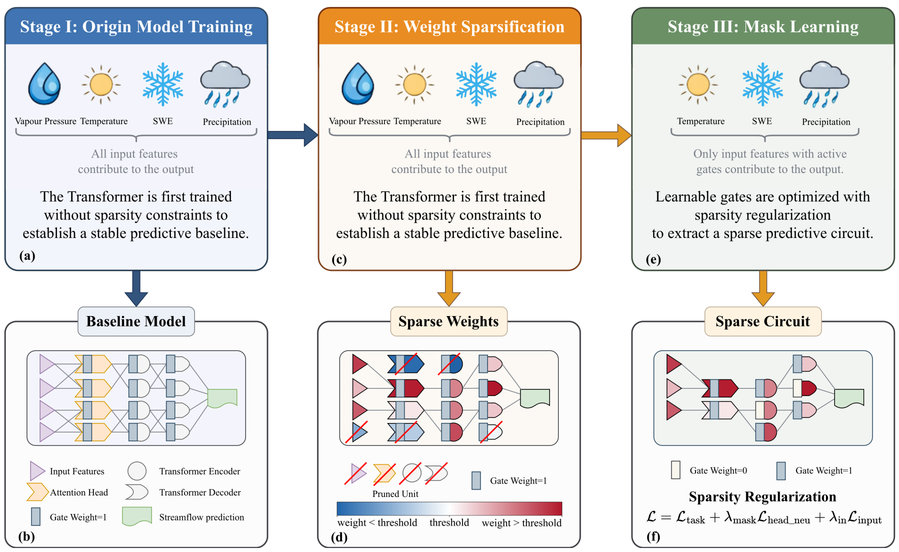

Fig. 7 Three-stage sparsification framework for predictive circuit discovery in Transformer. The model is trained through three stages to progressively transform a dense Transformer into a sparse predictive circuit. (a–b) Stage I: Origin model training, where the Transformer is trained without sparsity constraints and all input features contribute to the prediction. (c–d) Stage II: Weight sparsification progressively prunes redundant connections according to the sparsity schedule. (e–f) Stage III: Mask learning optimizes learnable gates so that only a subset of input features contributes to the final prediction. For clarity, only a subset of connections between layers is illustrated in the figure, while the actual model remains fully connected, and decoder attention layers are omitted for simplicity. Note that the proposed method does not explicitly modify the network architecture; sparsity is achieved by setting weights or gates to zero, while the visual removal of units in the figure is only for illustrative purposes. The number of input variables shown in the figure is fewer than the actual input feature set; this simplification is adopted solely for visual clarity.

To enhance the interpretability of the model, we introduce a structured gating mechanism to filter input variables, attention heads, and feed-forward neurons[68]. Through these gating mechanisms, we can selectively retain input features or parts of the network structure, thus enabling more efficient computation (Fig. 7(e–f)). By controlling the activation of gates, we are able to gradually filter out the features and pathways that have the most significant impact on the prediction. This structured sparsification approach helps the model focus on the most important inputs and computations, improving computational efficiency and enhancing model interpretability[69].

To avoid prematurely removing potentially important computational pathways, our training process is divided into three stages[34].:

Stage I (Origin Model Training): The model is first trained without sparsity constraints to establish a stable predictive baseline. In this stage, all input features and computational pathways are retained to ensure the model has sufficient expressive capacity (Fig. 7(a–b)).

Stage II (Weight Sparsification): In this stage, weight sparsification is gradually imposed, reducing redundant connections in the network. We use an annealed sparsity schedule to progressively increase the sparsity ratio, controlling which connections are pruned[70]. The process sets a target sparsity ratio (from the initial $\rho_{\max}$ to the final $\rho_{\min}$) to progressively reduce the complexity of the model (S2(c–d)).

Stage III (Mask Learning): Finally, learnable gates are optimized to further refine the network (Fig. S2(e–f)). We regularize the activation probabilities of the gates, guiding the network to select the most important input features and computational pathways[71]. The goal is to make the computational pathways of the network more sparse by adjusting the gate activation probabilities, improving the model's inference efficiency.

These stages combined result in a compact predictive circuit that provides efficient and accurate predictions with fewer computational resources[33]. Subsequently, we conduct an interpretability analysis of the model across multiple dimensions and evaluation metrics. If you are interested in the specific methods, please refer to the Supplementary Information 2 for further details (e.g., formal model definitions, interpretability techniques, etc.).

4.3 Data

This study uses the CAMELS-US dataset (https://doi.org/10.1016/0022-1694(89)90184-4), which provides daily hydro-meteorological time series for minimally disturbed catchments across the contiguous United States[72]. We select this dataset for three reasons: (1) catchments are weakly affected by direct human regulation, which helps identify more robust hydrological patterns; (2) long and standardized daily forcing–streamflow records support deep learning time-series training and cross-basin comparison[44]; and (3) the data organization aligns with large-sample hydrology workflows, enabling reproducible evaluation[24]. To ensure temporal consistency across stations, we use a common analysis period from 1980-01-01 to 2014-12-31. Based on this period, a total of 521 stations are retained after data quality control and screening.The model input variables consist of meteorological forcing and antecedent hydrological information. Specifically, the meteorological forcing includes multiple daily atmospheric variables, while the antecedent hydrological information is represented by historical streamflow sequences to capture catchment state memory. Additional details can be found in the Supplementary Information 2.4.

The prediction task is formulated as a sequence-to-one regression problem, where a sliding window of length $L = 15$ days is used to predict next-day streamflow. All features are standardized using training-set statistics, and the data are split chronologically into training (70%), validation (15%), and test (15%) subsets.

The model is implemented in PyTorch (Python 3.10) and trained on a CUDA-enabled Linux server. A three-stage pipeline (dense pre-training, weight sparsification, and mask learning) is optimized using AdamW with weight decay. Hyperparameters are tuned via Bayesian optimization[73] with validation NSE as the objective, and the selected configuration is consistently used across all stages and subsequent explainability analyses. Further details are provided in Supplementary Information 4.1.

Data availability

The data that support the findings of this study are publicly available. The streamflow and meteorological data are derived from the CAMELS-US dataset[72]. The dataset is accessible via: https://doi.org/10.1016/0022-1694(89)90184-4.

Code availability

The source code and processed datasets are available in the GitHub repository: https://github.com/Polyethylenekjc/Sparse-Transformer-In-Streamflow-Prediction.

CRediT authorship contribution statement

Jiachen Kong: Conceptualization, Methodology, Software, Writing – original draft. **Zhaocai Wang**: Methodology, Software, Validation, Supervision, Writing – review & editing. **Jun Wang**: Validation, Supervision, Writing – review & editing. **Zhaoyang Zhu**: Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 5 Appendix 1: Main Terminology Overview

Full Name	Abbreviation	Description
Physics-Informed Neural Networks	PINNs	Optimization constraint, emphasizes that predictions must obey physical laws, e.g., streamflow predictions satisfy mass conservation.
Sparse Transformer for Explainable Hydrology	STEH	Architecture design, emphasizes sparse and analyzable internal paths, e.g., identifying paths responsible for evapotranspiration.
Circuit	Circuit	Functional subset of computational units selected by sparsification (e.g., specific heads/neurons) that jointly supports prediction.
Structural	Structural	Organization pattern of components and interactions within a circuit (e.g., layer-wise arrangement and dependency layout).
Information Flow Pathway	Pathway	Directed route of information propagation within a circuit (e.g., precipitation-to-runoff transfer through key nodes).
Explainable AI	XAI	Overall goal, broad transparency and interpretability, e.g., understanding why the model predicts more accurately during drought periods.

Appendix 2: Confusable Concepts and Verification Methods

Full Name	Verification / Implementation Method
Explainability	Post-hoc analysis (SHAP / LIME) or intrinsic structural analysis to understand model prediction behavior.
Physical Structure Explainability	Structural sparsification + linear probing to verify whether internal computational paths correspond to physical quantities (e.g., soil moisture).
Causal Contribution	Rigorous causal verification, including causal graphs / SEMs, intervention experiments (Do-Calculus), and counterfactual / A/B comparisons to ensure true causal effect from input to output.
Structural Causal Contribution	Structured intervention + causal pathway verification + counterfactual comparison, to verify that internal components (attention heads, paths) are necessary and irreplaceable in the input–output causal chain.

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